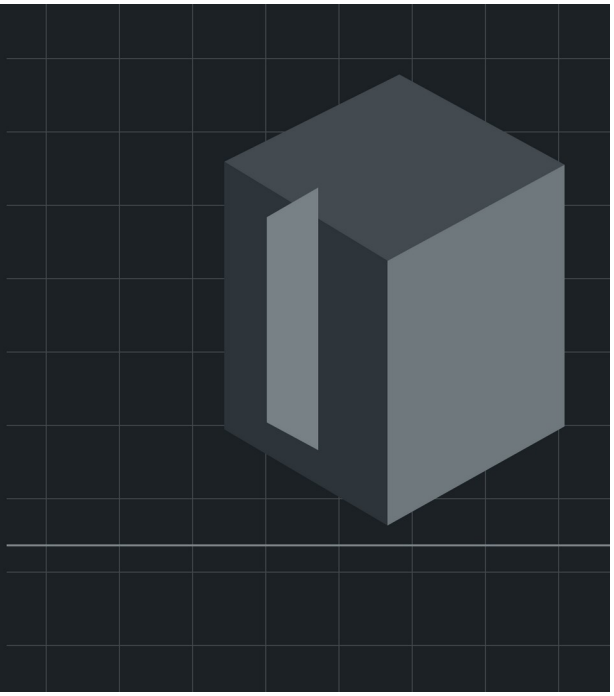
BOX1 / THE ENTERPRISE PLATFORM

BUILT TO EVOLVE.

LEGACY AT THE CORE.
MODERN AT THE EDGE.

THE CORNERSTONE OF THE ENTERPRISE

DOMAIN HELD SINCE 2002 / BOX1 OPERATIONS



BOX1 OPERATIONS

An AI-enabled enterprise application platform

An idea with history. A platform with purpose.

Box1 is a domain held since 2002 and the long-imagined foundation of a larger enterprise. Box1 Operations gives that idea a concrete first expression: a fictional operations business used to demonstrate what an integrated application platform can accomplish. This is not a collection of technology demos. It is one working system whose business transactions depend on modern applications, trusted delivery, infrastructure, legacy integration, and AI.

The guiding aesthetic mirrors the architecture: legacy yet modern. Retain what works. Make change deliberate. Use every platform capability to solve a problem that operators, developers, or customers can actually recognize.

THE BUSINESS

Distributed inventory, customer orders, equipment telemetry, and a VM-hosted ERP form a credible enterprise operating environment.

THE PLATFORM

OpenShift, service mesh, pipelines, GitOps, trusted supply chain, observability, virtualization, and local inference form one operating model.

The architecture: every capability has a job

A responsive operator console and customer portal sit above independent transactional and event-driven services. Orders reserve inventory; Kafka moves fulfillment and telemetry events; a legacy ERP remains the system of record. A governed Qwen assistant uses platform and application evidence to explain what happens when the workflow breaks.

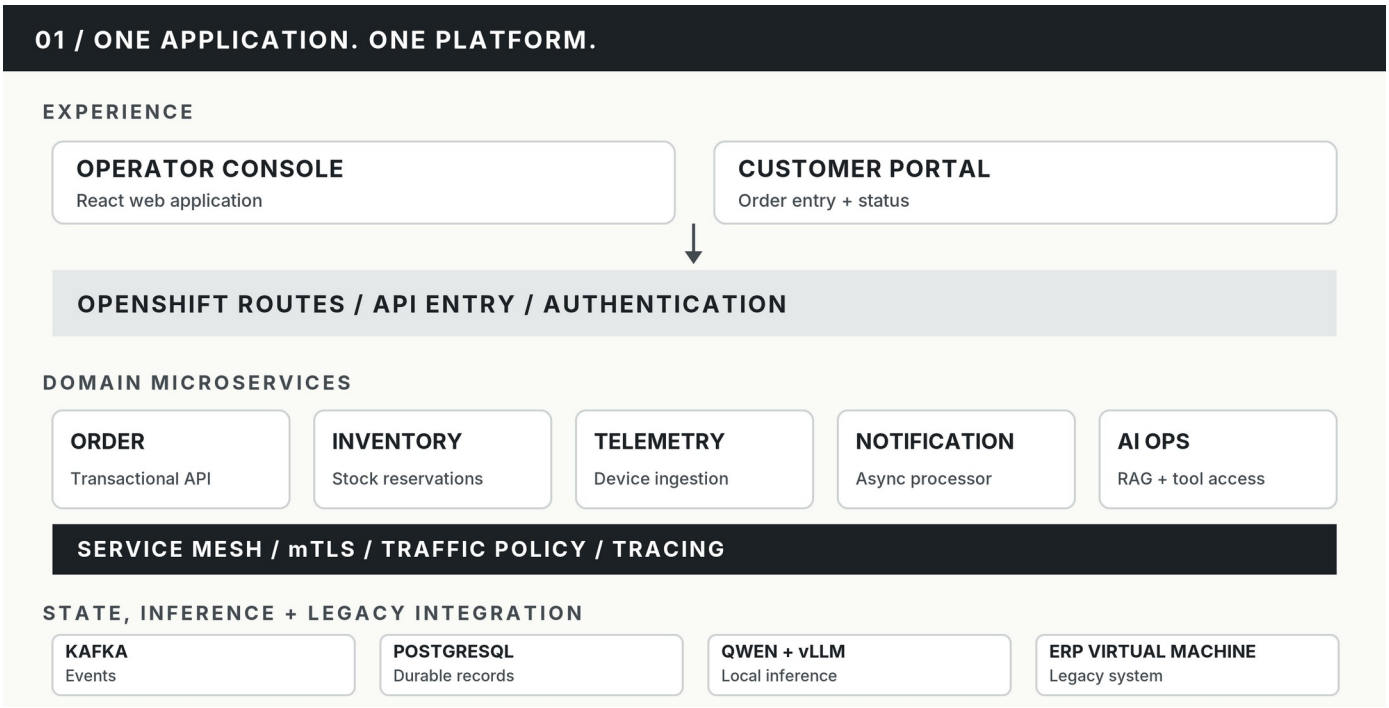


FIGURE 1 / The application and its platform foundation share one set of business transactions.

Built, secured, observed, and modernized

01 / APPLICATION SERVICES

Orders, inventory, telemetry, notification, and AI Ops run as independent microservices. Kafka connects asynchronous processes; PostgreSQL preserves durable business state.

02 / PLATFORM CONTROLS

Service Mesh governs identity and traffic. TSSC, Tekton, Quay, and GitOps establish a repeatable, inspectable software supply chain.

03 / OPERATIONAL INTELLIGENCE

OpenTelemetry, metrics, logs, traces, alerts, and SLOs connect user-facing incidents with the services and deployments responsible.

04 / HYBRID MODERNIZATION

OpenShift Virtualization keeps a RHEL-hosted ERP useful while APIs, event contracts, and newer services evolve around it.

AI is an operator, not a bypass. vLLM hosts local Qwen inference on the RTX 5090. The assistant may retrieve documentation and read approved telemetry; privileged changes require human approval, policy enforcement, and an audit trail.

From incident to insight

The flagship scenario deliberately slows an order between inventory reservation and the ERP acknowledgement. Instead of producing a generic alert, Box1 Operations connects the customer symptom, the transaction path, and the supporting evidence. The AI assistant summarizes what happened; an operator decides how to recover.

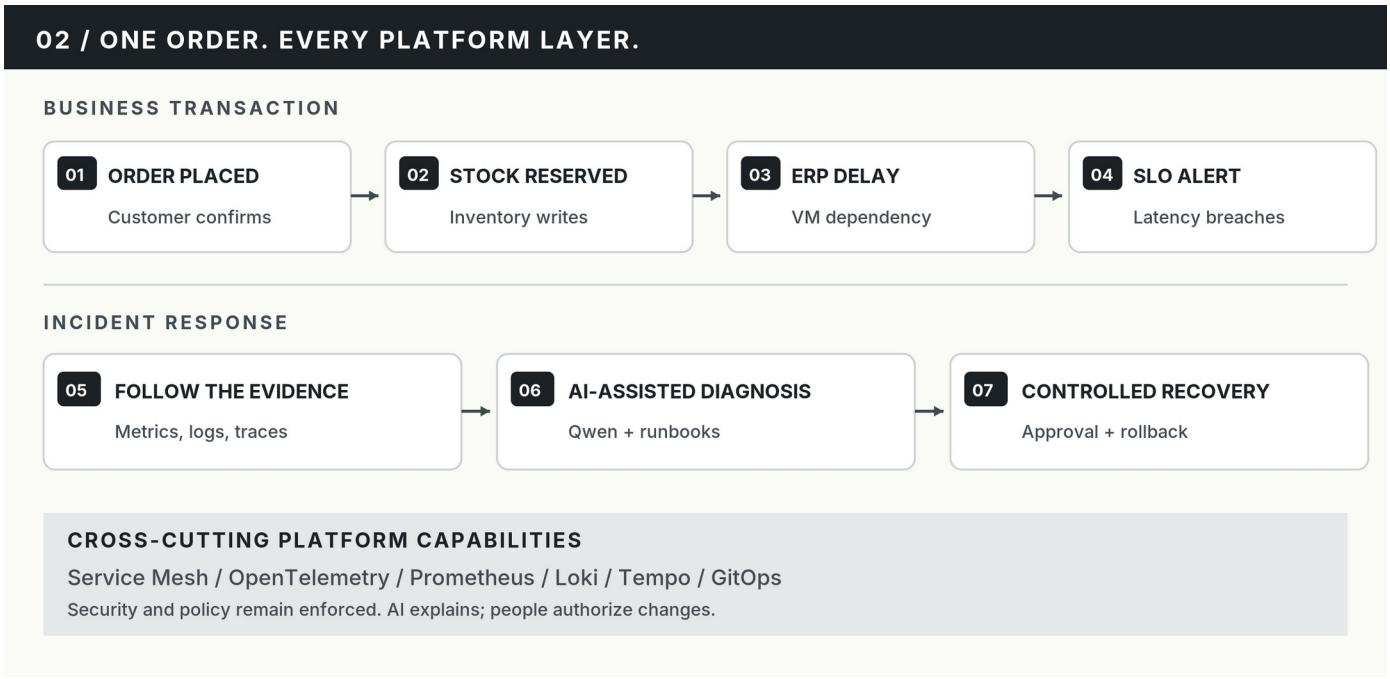


FIGURE 2 / Distinct transaction and response paths keep all labels readable and the causal story clear.

A practical sequence, not a technology checklist

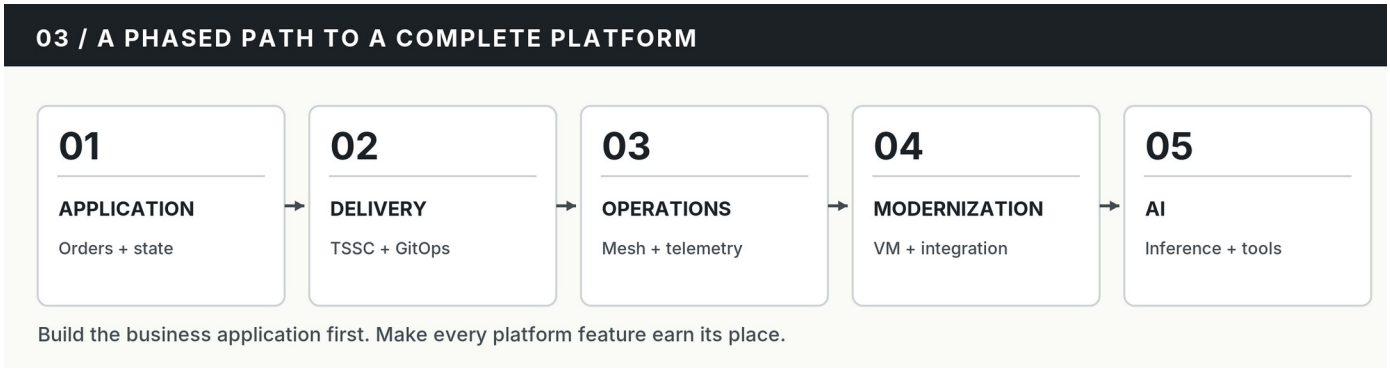


FIGURE 3 / Each phase introduces a capability only after the business workflow can demonstrate its value.

Definition of success

REPEATABLE + OBSERVABLE

Rebuild from Git. Place an order end to end. Inject a failure and follow it across APIs, Kafka, the service mesh, and the legacy VM.

SECURE + GOVERNED

Produce signed software artifacts. Demonstrate canary recovery. Let Qwen explain evidence while human approval gates remediation.

Box1 is intended to be a durable reference application, not a disposable lab exercise: a living expression of a domain and an enterprise ambition that began in 2002.